

High Bandwidth Memory

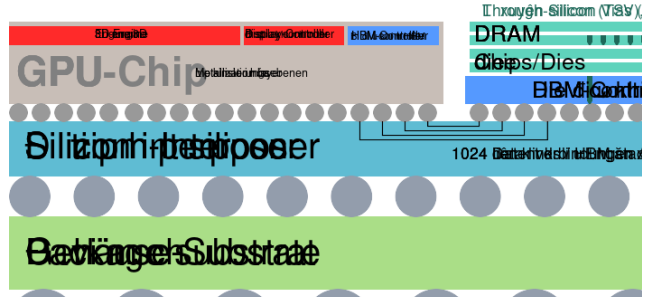
HBM stacks several DRAM dies into one memory package.

TSVs are copper paths that pass vertically through the silicon dies.

Microbumps form the electrical contacts between adjacent dies.

Thousands of short parallel connections provide a wide data interface and high bandwidth.

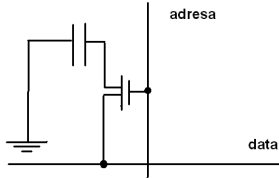
Source: Image: ScotXW, Wikimedia Commons, CC BY-SA 4.0. Cropped to the memory stack.



Vertically stacked DRAM dies connected by TSVs and microbumps

Electrical and TSV-related failures

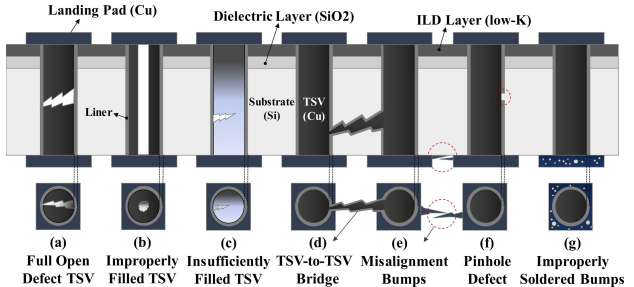
Electrical



DRAM electrical covers internal cell and circuit faults. Leakage can reduce the stored charge. Circuit defects can cause abnormal current, timing errors or failed reads.

dram_electrical

TSV-related



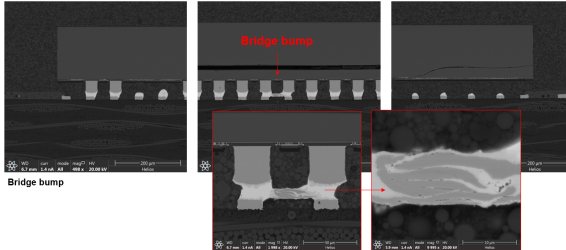
A **TSV open** breaks or weakens a vertical path. A **TSV short** creates an unintended connection to another path or the surrounding material.

tsv_open_short

Source: Images: Cuisinier2, Wikimedia Commons; Lee et al., PLOS ONE 14(8), 2019, Fig. 1, CC BY 4.0.

Microbump defects and die cracks

Microbump open or bridge



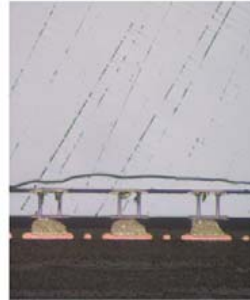
An **open** leaves an intended die-to-die contact disconnected. A **bridge** joins neighbouring contacts and creates a short.

microbump_open_bridge

Source: Images: SNP Tech, bridge-bump failure analysis; Orii et al., IMAPS 2013, Fig. 3a.

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Die crack

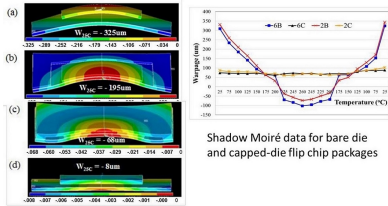


A fracture in a silicon die can cut circuitry or grow under mechanical and thermal stress.

die_crack

Warpage and interface defects

Warpage



Shadow Moiré data for bare die and capped-die flip chip packages

FEM results of warpage for comparing various package structures: (a) bare die; (b) with stiffener; (c) with lid; (d) capped-die

Uneven

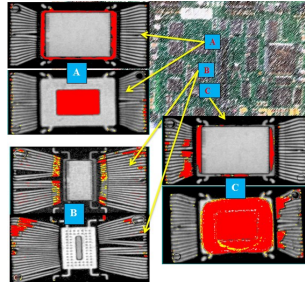
expansion and contraction bend a die, stack or package away from flatness. The deformation adds stress to bonded connections.

warpage

Source: Images: Lamar University, Electronic Packaging Research; Semiconductor Engineering, scanning acoustic microscopy examples.

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Underfill void and delamination



An **underfill void** is a

pocket where supporting material is missing.

Delamination is separation along an interface that should remain bonded.

underfill_void

delamination